LifeEngine: findings so far

Burrowing prey, claw cells, predators and soil · browser runs plus two headless batches (300 runs) · 18 October 2026

> **Main findings**
>
> 1. **Soil decides whether a world survives.** Most worlds that die do so early, when plants can't get through the crash that follows the first predator boom. Faster soil regrowth, denser soil and bigger fertile patches all make survival more likely. On sparse soil with small patches, every run died.
> 2. **Free claws keep burrowing prey alive.** When claw cells cost food, burrowing prey disappear in every run and on every soil layout. When claws are free, they are the main prey wherever a world survives.
> 3. **Free claws are exploited by predators.** Predators evolve mostly from the clawed prey (via the coin flip) and keep the claws. About 90% of predators carry claws they never use, and eyeless clawed predators can take over a world.
> 4. **Faster predators don't wipe out prey; they end up scarcer.** Predators that move every tick hold the fewest predators and survive less often than baseline across soil layouts.
> 5. **The seeded predators always die out.** In every batch run, all later predators evolved from prey or plants.

1. What was run

Set	Runs	What varied	Folder in `Observations/`
Early browser runs	2	before and after the tagging fix	`min burrow fertility 1`, `fixed tagging run`
Claw tracking (browser)	1	per-role claw use recorded	`claw tracking`
Claws cost food (browser)	1	claw benefits off	`claw dies out (not free)`
Claws free (browser)	1	claw benefits on	`claws free`
Batch 1 (headless)	30	6 configurations × 5 placement seeds, one soil layout	`claw-soil-predator`
Batch 2 (headless)	270	6 configurations × 9 soil layouts × 5 replicates, each with its own soil seed	`soil-layouts`

Both batches use the same six configurations. `baseline` matches the browser *claws free* run; each other configuration changes one thing from it:

Config	Change from baseline
baseline	none: claws free, no ageing while hidden, stay hidden near predators, soil regen chance 0.05%, predators move every 3 ticks
claw-off	all three claw benefits off
soil-slow / soil-fast	soil regen chance 0.01% / 0.2%
pred-2 / pred-1	predators move every 2 ticks / every tick

- **Every run starts with** 10 predators, 10 prey, 10 clawed prey and 39 plants on a 384 × 193 grid (cell size 5), and lasts up to 250,000 ticks.
- **Only the starting layout is seeded** (soil seed and placement seed). Each run is otherwise random, so replicates differ in how they play out.
- **Batch 1** uses one soil layout (seed 42, density 0, resolution 20, 55% fertile). **Batch 2** adds 3 soil densities (density threshold −0.1, 0, 0.1: about 68%, 50% and 33% fertile) × 3 patch sizes (resolution 10, 20, 40: big to small patches), and pairs each replicate with its own soil seed (42, 7, 123, 2024, 99).

Changes to the simulator along the way

- **Tagging:** organisms are tagged predator (killer cell, no producer) or prey (anything else that isn't a plant) from their body, so evolved prey and predators are recognised. A clawed organism gaining a killer cell has a set chance of becoming a predator (the coin flip, 0.5 here).
- **Species:** an offspring that switches between predator and non-predator founds a new species, whatever its body.
- **Claw tracking:** logs count, per role, organisms with claw cells, those that have ever burrowed, and those underground.
- **Run Setup** (`src/EnvironmentParams.js`): seeded soil and placement, organism counts, soil regrowth, the coin flip and the claw benefits, all recorded in each run's config.
- **Headless runner and batch launcher** (`headless/`): the same simulation code without drawing, several runs at a time. Both batches together took about 2 hours 50 minutes.

2. Browser runs

Before and after the tagging fix

- Before the fix, evolved predators weren't tagged. Prey couldn't recognise them as predators, and organisms evolving from plants that moved weren't prey that predators could eat. Predators starved, and prey were then killed by killers they didn't treat as predators.
- After the fix, the population plots looked more normal. But claw cells were common even in organisms that never burrowed. This led to tracking claw use by role.

Claw tracking: burrowing prey crash from inside their own lineage

- Burrowing prey were the main prey for about the first 65k ticks, and were burrowing right up to their crashes at about 67–70k and 148–150k.
- The predators that overexploited them had evolved from the burrowing prey's own lineage, via the coin flip.
- Most predators and many plants carried claws they never used, because claws were free.

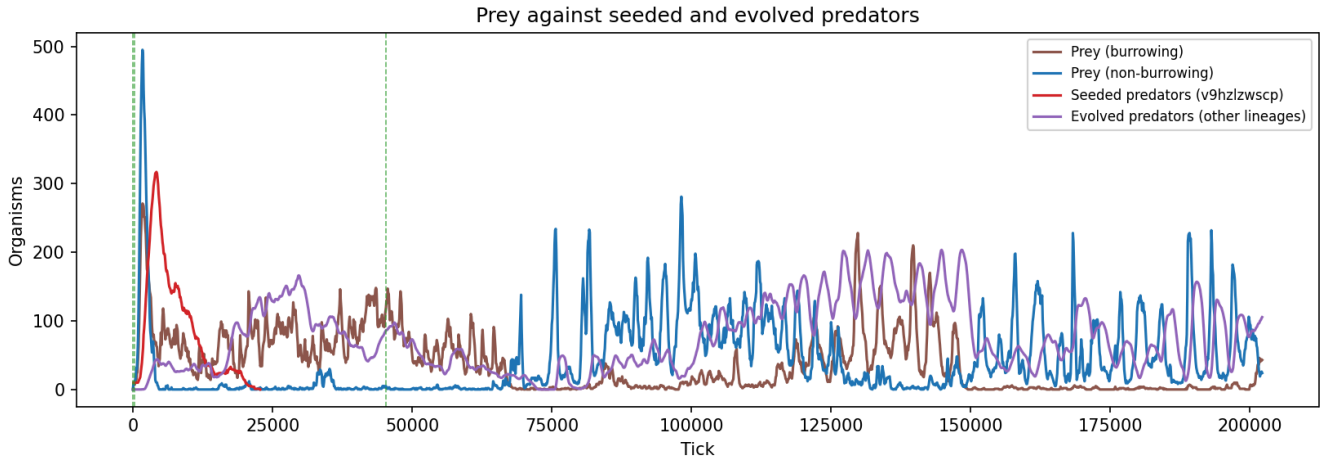


Figure 1. Claw tracking run: burrowing prey (brown) dominate early, then decline as evolved predators (purple) rise.

Claws cost food: burrowing prey die out within about 4,000 ticks

- With all three claw benefits off, the 10 seeded predators grew to about 270 by tick 4,000 on the first prey boom. All prey crashed: non-burrowing prey fell by about 90%, and burrowing prey died out.
- Burrowing prey bred at about half the rate of other prey: their claw adds to the food they need, they can't eat underground, and they age while hidden. Starting from a much smaller population (peak 62 against about 400), they had nothing left to recover from.
- This wasn't evolutionary suicide: a predator boom overran every prey type, and the smallest, slowest-breeding group didn't come back.

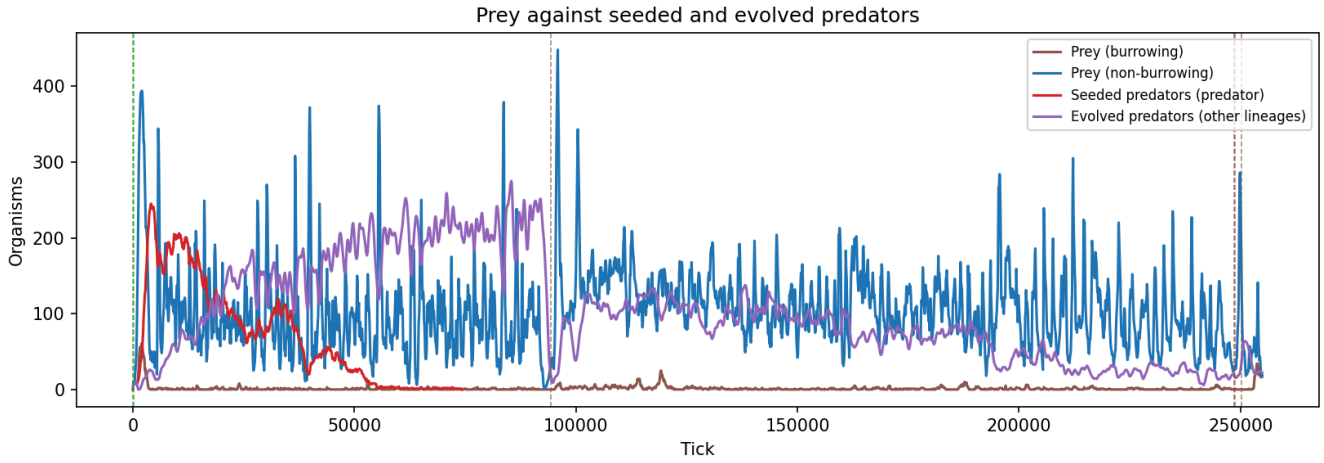


Figure 2. Claws cost food: burrowing prey (brown) vanish early; non-burrowing prey (blue) carry on against seeded (red) and evolved (purple) predators.

Claws free: burrowing prey persist, then eyeless clawed predators take over

- Burrowing prey were the main prey from about 20k to 110k ticks, with 35–70% of them having burrowed.
- The dominant predators all came from the clawed-prey lineage. From about 80k, 80–97% of predators carried claws, and none ever burrowed.
- The most successful predator had a mover, a killer and 3 claws, with no eye and no mouth. Free claws mean it pays food for only 2 cells (4 food, against 11 for the seeded predator), while every cell, claws included, counts toward lifespan and health. Mouths are useless to predators, which only feed by killing, and eyes cost food. A cheap random walker that breeds about 3× as often outcompetes a hunter.

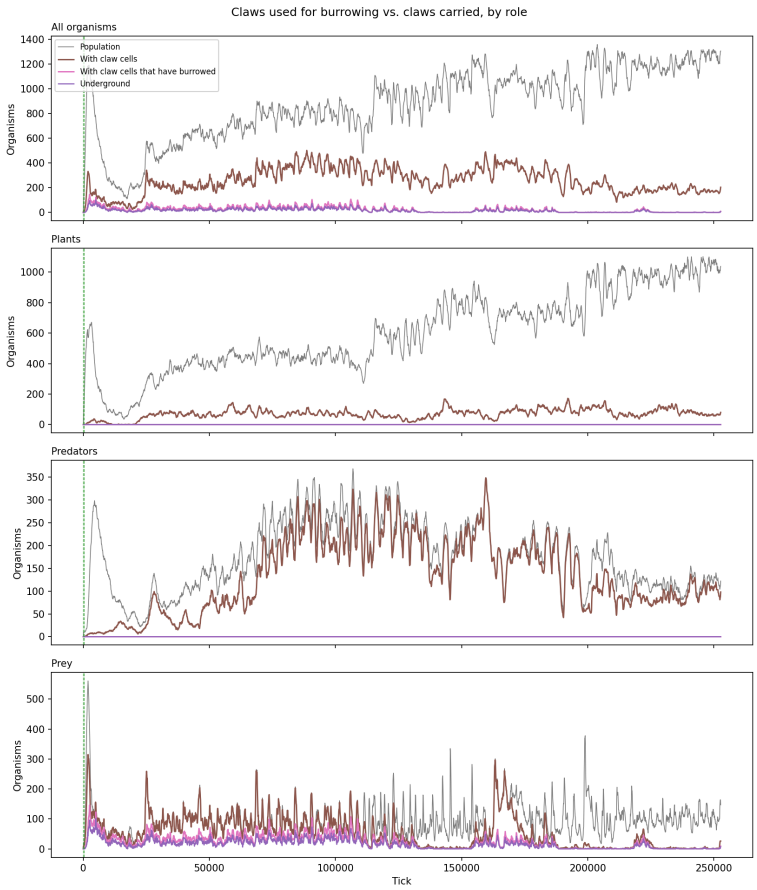


Figure 3. Claws free: claws carried (brown) against claws used for burrowing (pink) and underground (purple), by role. Almost every predator carries claws, but none burrow.

3. Batch 1: one soil layout, six configurations (30 runs)

Config	Lasted 250k	Burrowing prey	Non-burrowing prey	Predators	Plants
soil-fast	**5/5**	**132**	6	160	527
baseline	3/5	78	4	185	331
claw-off	3/5	**2**	**98**	87	**786**
pred-2	3/5	54	9	61	334
pred-1	3/5	20	26	41	330
soil-slow	**0/5**	47	1	21	173

Populations are means over ticks 100k–250k for runs still going (soil-slow: a single run there).

- **Why worlds die early:** predators wipe out the first prey boom by about tick 4k. Uneaten food then piles up (about 1k → 5k cells) and blocks plant offspring from being placed, while burrowing wears down about 20% of the fertile soil, which takes about 14k ticks to recover at the default rate. The browser *claws free* run nearly died the same way (35 plants at 16k).
- **Seeded predators died out in every run,** between 7k and 87k ticks. With claws free, 94% of later predators (baseline and soil-fast) came from the clawed-prey lineage; with claw-off, about half did.
- **Eyeless clawed predators came back on their own** in 3 runs (82–98% of predators), never with claw-off. Free claws also made bodies bigger (about 6.2 cells against 4.7).
- **pred-1** had the highest mutation rate (12.7% against about 8–9%) and the most variation between runs.

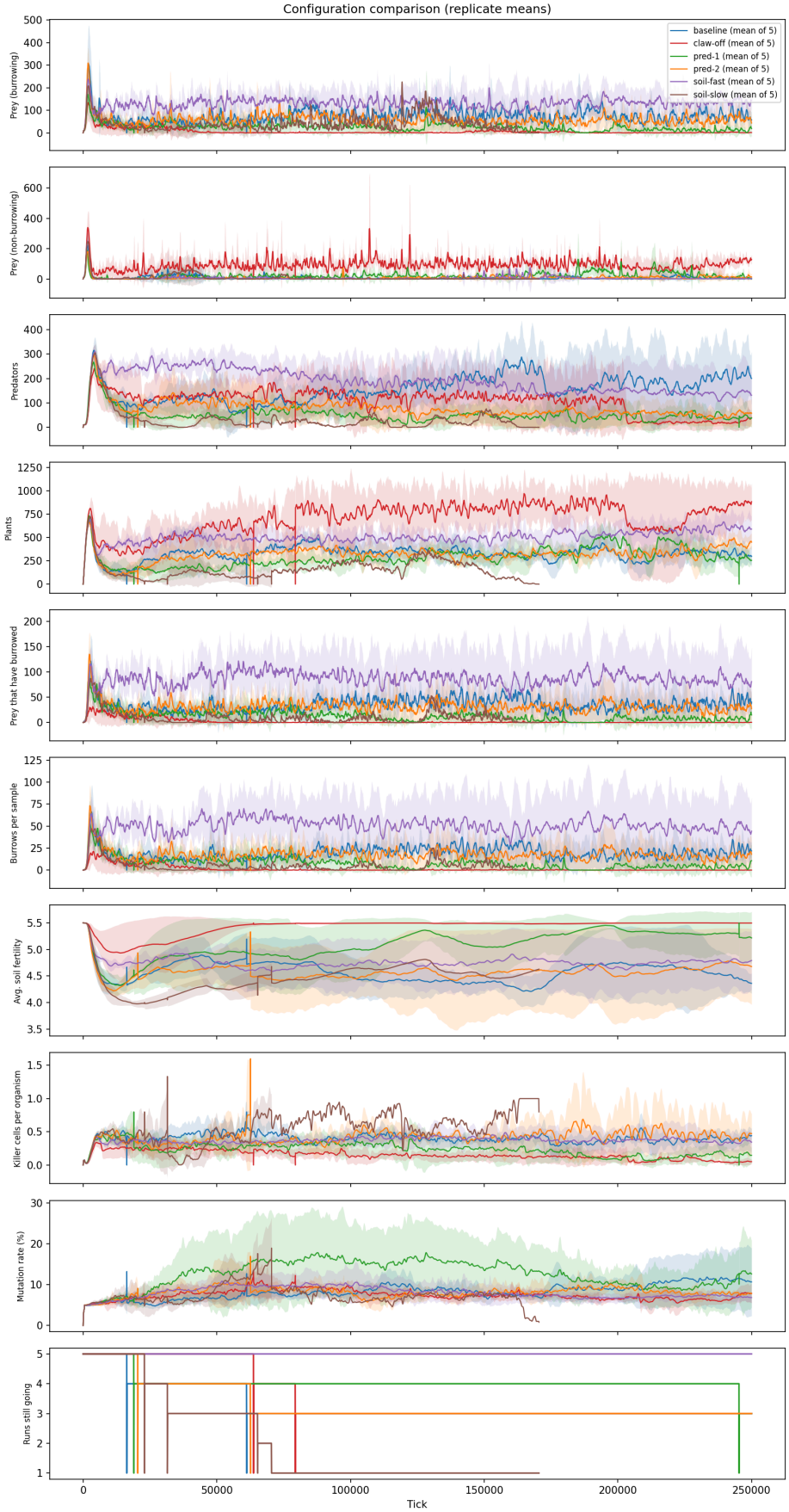


Figure 4. Batch 1: replicate means (± 1 std) per configuration. The last panel shows how many runs are still going; means after a run dies are over the runs still going.

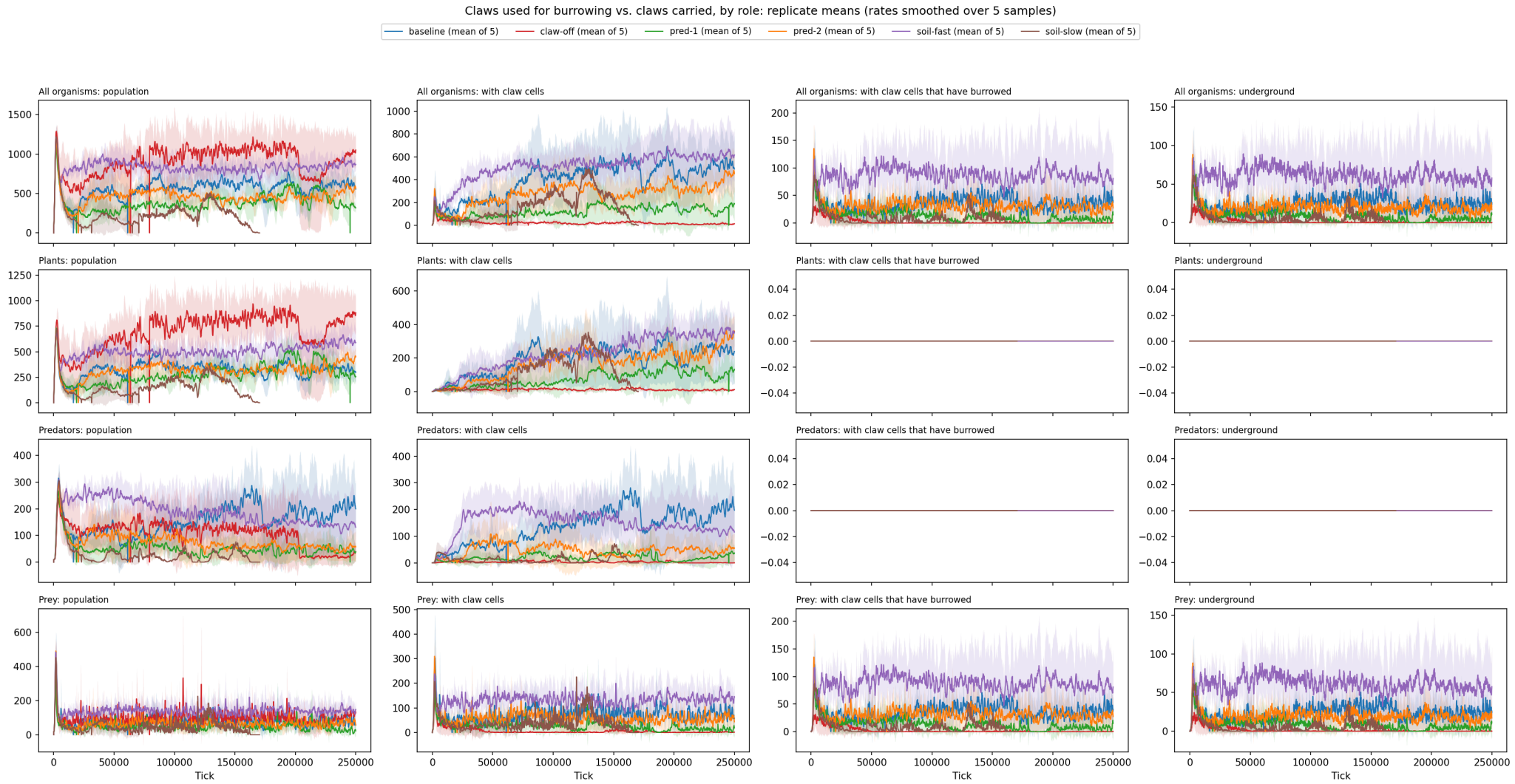


Figure 5. Batch 1: claws carried against claws used for burrowing, by role. With claws free, predators and plants accumulate claws they never use.

4. Batch 2: nine soil layouts (270 runs)

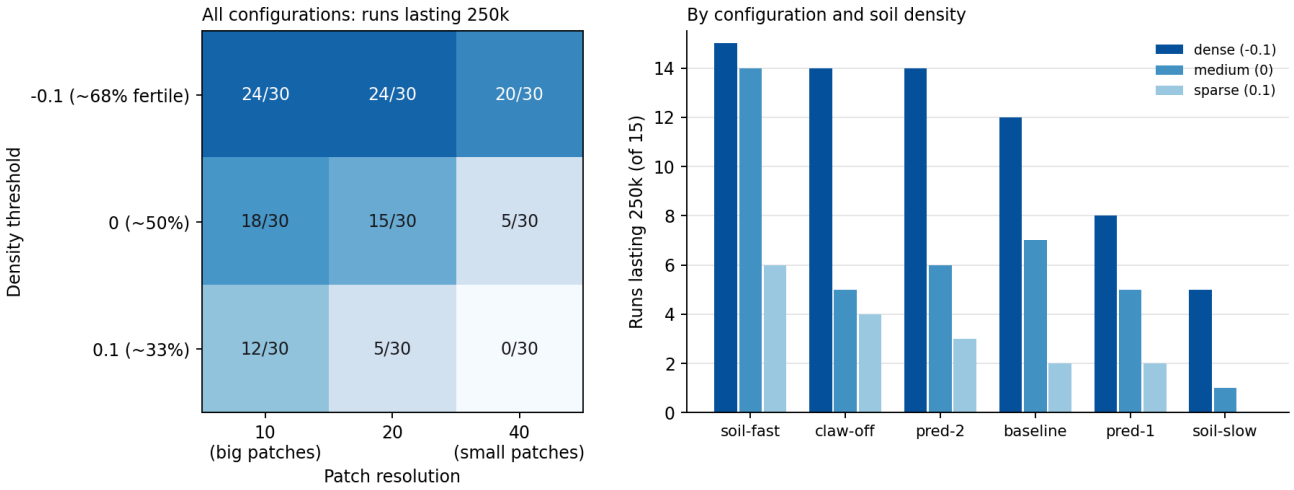


Figure 6. Batch 2 survival. Left: all six configurations by soil density and patch size. Right: each configuration by soil density (all patch sizes).

- **Density matters most:** 68/90 runs survived on dense soil, 38/90 on medium and 17/90 on sparse.
- **Small, scattered patches make it worse:** on sparse soil with small patches, all 30 runs died. On sparse soil, big patches kept some runs alive in every configuration except soil-slow, likely because large connected fertile areas let plants get through the early crash.
- **The layout itself matters too:** soil seed 123 survived 18 of 54 runs, against 24–28 for the other seeds. Patch resolution 10 varies a lot between seeds (44–76% fertile at density 0).
- **Most deaths are early:** 84 of 147 before tick 30k (median about 25.6k), the same early plant crash as in batch 1.

How each configuration fares across layouts

Config	Lasted 250k (of 45)	Late-run pattern (late = ticks 100k–250k, runs that died count as 0)
soil-fast	**35**	most predators and most burrowing prey on almost every layout (up to 193 burrowing prey); fails on sparse soil with small patches
claw-off	23	burrowing prey about 0 everywhere; non-burrowing prey and plants (up to about 970) highest; survives dense soil with small patches better than baseline (5/5 against 2/5)
pred-2	23	about as well as baseline, or better
baseline	21	burrowing prey grow with soil fertility: about 21 on sparse soil, up to 92 on dense
pred-1	15	worst apart from soil-slow, even on dense soil (8/15); fewest predators
soil-slow	6	survives only on dense or medium soil with big patches, and rarely

- **Claws on predators hold on every layout:** late in the runs, about 86–93% of predators carry claws in every configuration with free claws (77% in soil-slow), against 11% with claw-off.
- **pred-1 changes the batch 1 conclusion:** on the single layout it survived as often as baseline; across layouts it is clearly less robust.

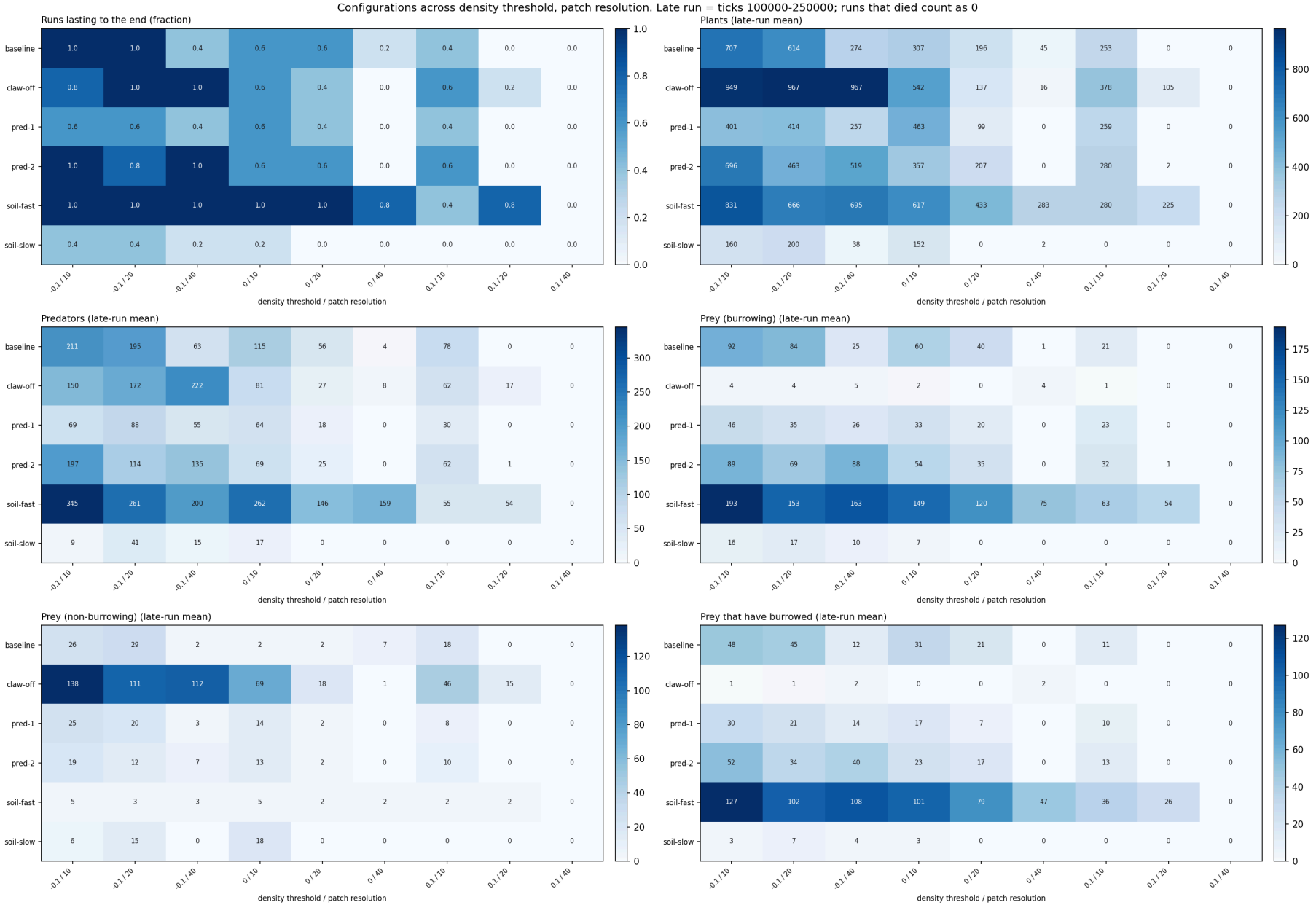


Figure 7. Batch 2: every configuration (rows) against every soil layout (columns, density threshold / patch resolution). Top left: fraction of runs lasting 250k. The rest: late-run means (ticks 100k–250k), with runs that died counting as 0.

5. Putting it together

- **The system's weak point is the early plant crash.** It appears in the browser runs and both batches, and soil (how fast it regrows, how dense it is, how connected the patches are) decides whether a world gets through it. This makes survival noisy and affects every other comparison.
- **Burrowing works only when claws are free**, and then its benefit grows with soil fertility. When claws cost food, burrowing prey never establish.
- **The claws-free rule is a loophole for predators.** Claws lower the food needed to reproduce while still adding lifespan and health, so predators (mostly descended from clawed prey) keep them, and minimal, eyeless clawed predators can win.
- **Predator speed doesn't give the predicted pressure.** Faster predators overexploit and stay scarce rather than forcing more burrowing; burrowing prey are fewer, not more, with pred-1.

Caveats

- 5 replicates per configuration and layout; individual cells of the batch 2 heatmaps rest on 5 runs.
- Batch 1 late-run means cover only runs still going; batch 2 heatmaps count dead runs as 0. The two are not directly comparable.
- The eyeless and clawed predator shares in batch 1 are estimated from each species' founder body.
- The browser runs are single runs on one layout.

Next steps

1. **Free claws for prey only.** Rerun baseline and soil-fast to test whether the clawed and eyeless predators come from the loophole.
2. **Remove the early plant crash** (e.g. `foodBlocksReproduction` off, or faster soil regrowth as the base), so comparisons don't depend on whether a world happens to survive.
3. **Change only the coin flip** (`clawPredatorChance` 0 against 0.5) to separate the clawed-prey-to-predator route from the claw cost.
4. **Use dense or medium soil with big or medium patches** for future comparisons, where most configurations survive, and keep several soil seeds per configuration.